Oxford Cambridge and RSA Examinations

GCE

Economics

H460/02: Macroeconomics

A Level

Mark Scheme — Predicted Paper 2 (Advanced)

MARKING INSTRUCTIONS

Marks awarded must relate directly to the marking criteria. Alternative correct answers and unexpected approaches must be credited where they show relevance. Mark schemes should be read in conjunction with the published question paper.

For answers marked by levels of response:

- To determine the level — start at the highest level and work down until you reach the level that matches the answer.
- To determine the mark within the level, consider whether the answer is at the borderline (bottom), just enough (above bottom/middle), meets criteria with slight inconsistency (above middle), or consistently meets the criteria (top).
- For extended-response questions marked with (*), the side of the argument presented first should be credited as Analysis; the other side is credited as Evaluation.

Question 1 (a)

Answer	Mark	Guidance
Using information from the stimulus material, identify two causes of Argentina's currency depreciation. Two from: • High / hyper-inflation (above 200%) eroding the real value of the peso (1) • Sovereign debt defaults / loss of investor confidence (1) • Capital flight / households and firms selling pesos for US dollars (1) • Rapid expansion of the money supply to finance government deficits (1) • Successive currency crises (1) • Negative real interest rates reducing demand for peso-denominated assets (1)	2	Annotate with ✓ Do not credit generic factors not referenced in the stimulus (e.g. "trade war", "poor exports"). Credit the first two responses given.

Question 1 (b)

Answer	Mark	Guidance
Calculate to one decimal place Argentina's real interest rate when its policy interest rate was 133% and inflation was 211%. **Correct answer = −78 or −78.0 (%)**	1	Annotate with ✓ Real interest rate ≈ Nominal rate − Inflation rate = 133 − 211 = −78 = −78.0 (to 1 d.p.) Do not credit 78 or 78.0 (sign required). Do not credit 0.78 or −0.78.

Question 1 (c)

Answer	Mark	Guidance
Explain whether the relationship shown in Fig. 1 between currency depreciation and inflation is the expected one. • Yes — there is a strong positive relationship between currency depreciation and inflation (1) • This is expected because depreciation raises the domestic price of imports (imported inflation); purchasing power parity (PPP) theory predicts that high-inflation countries experience offsetting currency depreciation (1) • Supporting evidence e.g. Argentina (74.2% depreciation / 94.7% inflation) and Turkey (42.8% / 41.6%) show both high depreciation and high inflation; Mexico (2.1% / 5.4%), Indonesia (3.4% / 3.6%) and India (3.0% / 5.6%) all show both low depreciation and low inflation (1) • Exception / limitation: in Argentina, inflation (94.7%) is substantially higher than depreciation (74.2%), suggesting domestic inflationary pressures (monetary financing of deficits) beyond what imported inflation alone can explain; or Mexico has higher inflation (5.4%) than depreciation (2.1%), indicating inflation is driven by domestic factors (1)	3	Annotate with ✓ Relationship — 1 Why expected — 1 Supporting evidence — 1 Exception — 1 Evidence must explain why it shows a relationship or exception rather than just listing a country.

Question 1 (d)

Answer	Mark	Guidance
Using Table 1, explain the relationship between real GDP growth and unemployment in IMF programme countries. • There is no clear / consistent relationship between real GDP growth and unemployment (1) • Evidence e.g. Kenya has the highest real GDP growth (5.6%) and a relatively low unemployment rate (5.7%), and Ghana has positive growth (2.9%) with the lowest unemployment (3.9%), suggesting a negative relationship; but Sri Lanka has the worst growth (−2.3%) yet the lowest unemployment of the contracting economies (4.7%); and Pakistan has slight contraction (−0.2%) with the highest unemployment (8.5%) (1) • Economic theory (Okun's Law) predicts an inverse relationship because higher growth raises derived demand for labour. This is partly observed but the data is mixed — the relationship is weak (1) • Exception / limitation: unemployment statistics differ between countries in how they are measured — large informal sectors in emerging economies can mask underemployment; labour market flexibility and population growth also affect the unemployment rate independently of real GDP growth (1)	4	Annotate with ✔ Relationship — 1 Evidence — 1 Explanation — 1 Exception / limitation — 1

Question 1 (e)*   Evaluate whether a depreciation of the Argentine peso is likely to improve the country's current account on the balance of payments. [8]

Level / Mark	Descriptor
Level 2 (5–8)	Good KU, good application with data, strong analysis (well-developed chains of reasoning), strong evaluation with a logical and well-supported judgement.
Level 1 (1–4)	Reasonable KU, reasonable application (possibly inconsistent), reasonable analysis in single links, limited evaluation with a stated or absent judgement.
0 marks	Response is not worthy of credit

Indicative content — Question 1(e)*

Knowledge and understanding

- The current account records trade in goods and services, primary and secondary income.
- Depreciation is a fall in the value of a currency under a floating exchange rate system.
- The Marshall–Lerner condition: depreciation improves the current account if $|PEDx| + |PEDm| > 1$.
- The J-curve: in the short run the current account may worsen before improving.

Application — context

- The Argentine peso depreciated from about 30 pesos/USD in 2018 to over 800 pesos/USD by December 2023.
- Argentina's exports are highly concentrated in primary commodities (soybeans, beef, maize).
- Argentina's export price index was 152.4 and import price index was 168.0 in 2023.
- Argentina has capital controls creating a black-market exchange rate at a premium to the official rate.

Analysis — depreciation likely to improve the current account

- A weaker peso makes Argentine exports (in USD terms) cheaper for foreign buyers, raising export demand and volumes.
- Imports become more expensive in peso terms, reducing import demand and import volumes.
- If the Marshall–Lerner condition holds, net exports rise, improving the current account.
- Argentina's export sectors (soybeans, beef, maize) may gain international price competitiveness, supporting foreign currency inflows.
- An improving trade balance can help Argentina rebuild foreign currency reserves — a key IMF condition.

Evaluation — depreciation may not improve the current account

- Argentina's exports are concentrated in primary commodities whose prices are set on world markets in USD — depreciation does not change the world price, only the peso value received by exporters.
- Demand for commodities is relatively price-inelastic, so the volume response may be small — Marshall–Lerner may not hold.
- Argentina relies on imported capital goods, fuel and intermediate inputs; more expensive imports raise production costs, eroding export competitiveness (imported cost-push inflation).
- The J-curve effect means the current account can worsen in the short run before any improvement.
- Capital controls distort the effective exchange rate — the black-market rate gap means official depreciation does not fully translate into competitiveness gains.
- Very high inflation (211%) can rapidly erase the competitiveness gained from nominal depreciation (real effective exchange rate may barely change).

Judgement may include

- The effect depends critically on whether the Marshall–Lerner condition holds for Argentina's trade composition.
- In the short run the current account may worsen (J-curve); improvements may only appear in the medium term.

- Depreciation must not be eroded by domestic inflation — without monetary and fiscal reform, real gains will evaporate.
- Commodity concentration limits the expenditure-switching effect of depreciation.
- Depreciation alone cannot fix structural problems — complementary supply-side reform is required.

Question 1 (f)*   Evaluate whether IMF loans are likely to benefit developing economies such as Argentina. [12]

Level / Mark	Descriptor
Level 3 (9–12)	Good KU and application, strong analysis (well-developed chains of reasoning), strong evaluation with a logical and well-supported judgement.
Level 2 (5–8)	Good KU and application, reasonable analysis (single links), reasonable evaluation with an unsupported judgement.
Level 1 (1–4)	Reasonable KU, reasonable application, limited analysis and limited evaluation.
0 marks	Response is not worthy of credit

Indicative content — Question 1(f)*

Knowledge and understanding

- The International Monetary Fund (IMF) provides financial assistance to member countries facing balance of payments or currency crises.
- IMF loans typically come with conditions — fiscal consolidation, structural reform and rebuilding of foreign currency reserves.
- The conditions associated with IMF lending are often described as "austerity" measures.

Application — context

- Argentina received the largest single IMF loan in the Fund's history (US$57 billion) in 2018.
- Argentina agreed a refinancing programme with the IMF in 2022.
- Over 40% of Argentina's population were living below the national poverty line by 2023.
- Other IMF programme countries include Pakistan (growth −0.2%, unemployment 8.5%), Sri Lanka (growth −2.3%), Ghana (growth 2.9%), Egypt (growth 3.8%) and Kenya (growth 5.6%) — outcomes vary widely.

Analysis — IMF loans benefit developing economies

- IMF loans provide a financial lifeline that may prevent uncontrolled default and total economic collapse.
- Access to IMF funds allows the central bank to rebuild foreign currency reserves, restoring confidence in the currency.
- The credibility of IMF conditions can lower bond yields by reassuring markets that fiscal discipline will be restored.
- Conditions requiring structural reform (reducing subsidies, improving tax collection, trade liberalisation) can address underlying causes of instability.
- The availability of an IMF programme can catalyse further private lending and foreign direct investment.
- Compared to uncontrolled default, the path through an IMF programme is typically less disruptive.

Evaluation — IMF loans may not benefit developing economies

- The austerity conditions (cuts to public spending, tax rises) deepen recession in the short term — Sri Lanka (−2.3%) and Pakistan (−0.2%) show contractions during IMF programmes.
- Cuts to food, fuel and transport subsidies disproportionately harm low-income households and worsen relative and absolute poverty.
- Loss of economic sovereignty — policy decisions are effectively made outside the country.
- Austerity-linked social unrest (as seen in Argentina, Greece) can destabilise political institutions.
- High-interest repayments commit future governments to long-run debt-servicing, potentially creating a cycle of dependency (Argentina has received repeated IMF packages).
- Conditionality may ignore country-specific context — "one-size-fits-all" reforms may not suit every economy.
- Kenya and Egypt have seen positive growth, suggesting success is not guaranteed by the programme alone but also depends on the recipient economy's structure.

Judgement may include

- The benefit depends heavily on the design of the programme and the starting position of the economy.
- In a severe balance of payments crisis, the alternative — uncontrolled default — is usually worse than an IMF programme.
- Outcomes differ widely: Kenya and Ghana show growth during IMF programmes, while Sri Lanka and Pakistan contracted.
- Short-run pain (austerity) may be worthwhile if the long-run gain (stability, growth, lower inflation) is secured.
- IMF loans are more likely to benefit economies that pair the loan with genuine structural reform rather than repeatedly refinancing.
- Argentina's pattern of repeated IMF programmes suggests the underlying causes of instability have not been addressed.

Section B — Question 2*

Evaluate, with the use of an appropriate diagram(s), whether quantitative tightening is an effective policy to reduce inflation. [25]

Level / Mark	Descriptor
Level 5 (21–25)	Strong KU, application, analysis and evaluation. Accurate, integrated diagram(s). Logical and well-supported judgement.
Level 4 (16–20)	Good KU, application, analysis and evaluation. Predominantly correct diagram(s). Unsupported judgement.
Level 3 (11–15)	Good KU, application, good analysis, reasonable evaluation. Unsupported judgement.
Level 2 (6–10)	Reasonable KU, application, analysis (single links) and evaluation.
Level 1 (1–5)	Limited KU, application, analysis and evaluation.
0 marks	Response is not worthy of credit.

Indicative content — Question 2*

Knowledge and understanding

- Quantitative tightening (QT) is the reversal of quantitative easing: a central bank reduces the size of its balance sheet by allowing bonds to mature without reinvestment, or by actively selling bonds.
- QT reduces the money supply and raises long-term interest rates.
- QT is a contractionary monetary policy instrument intended to reduce inflation.

Application — context

- The Bank of England, Federal Reserve and ECB have all conducted QT programmes since 2022 following the post-pandemic inflation surge.
- The Bank of England's balance sheet expanded from about £400bn to nearly £900bn through QE; QT is unwinding this.
- QT has coincided with policy-rate rises to complete the contractionary monetary stance.

Analysis — QT is effective in reducing inflation (with diagram)

- Diagram: AD shifts left as higher long-term interest rates reduce consumption and investment; the price level falls from PL1 to PL2 at a lower real GDP.
- Reducing the central bank's bond holdings withdraws liquidity from the banking system, raising the price of credit.
- Higher long-term yields reduce investment and interest-sensitive consumption (housing, durable goods), reducing AD and demand-pull inflation.
- QT signals a credible commitment to fighting inflation, anchoring inflation expectations and limiting second-round wage-price effects.
- QT reverses the wealth effects of QE — lower asset prices reduce consumption of wealth-holders.
- QT complements policy-rate rises by tightening financial conditions across the yield curve.

Evaluation — QT may not be effective

- QT works through long-term yields with lags of 12–24 months — the inflationary surge may have passed before QT has its full effect.
- Much of recent inflation was driven by supply-side shocks (energy, supply chains) — QT does not address cost-push inflation directly.
- QT can destabilise bond markets (e.g. UK gilt market turmoil in late 2022 required emergency BoE bond buying), undermining its credibility.
- Rapid QT can trigger financial instability — banks holding depreciated bonds may face capital pressure (cf. US regional bank failures in 2023).

- The effectiveness depends on the behaviour of commercial banks: they may reduce lending independent of central bank signals, making QT less predictable.
- QT coinciding with fiscal tightening risks an overly contractionary policy mix, harming growth.

Possible judgement

- QT is most effective against demand-pull inflation, less effective against cost-push inflation.
- The pace of QT matters — gradual unwinding avoids bond market disruption but may be too slow for urgent inflation control.
- QT is one part of a policy package — policy-rate rises, exchange-rate effects and fiscal policy all matter.
- In the current cycle, QT has probably contributed to falling inflation but the supply-side normalisation (energy prices falling) has also played a major role.
- Policy effectiveness depends on credibility — central banks with strong independence (Fed, BoE, ECB) have greater effect than those facing fiscal dominance (e.g. Argentina's BCRA).

Section B — Question 3*

Evaluate, with the use of an appropriate diagram(s), whether high levels of government debt necessarily harm an economy's long-run economic growth. [25]

Level descriptors as for Q2.*

Indicative content — Question 3*

Knowledge and understanding

- Government debt is the accumulated stock of past deficits expressed as a percentage of GDP.
- Long-run economic growth depends on increases in productive capacity (LRAS).
- Crowding-out: government borrowing can raise interest rates and displace private investment.

Application — context

- Japan's government debt is around 252% of GDP, the highest among major economies, yet bond yields remain low and growth modest.
- UK, US and eurozone debt-to-GDP ratios have risen substantially since 2008 and 2020.
- Developing economies such as Argentina, Sri Lanka and Ghana have all experienced debt crises recently.

Analysis — high debt harms long-run growth

- Diagram: LRAS may fail to shift right if investment is crowded out by government borrowing; growth stagnates.
- Higher debt-servicing costs crowd out productive government spending (infrastructure, education, healthcare).
- Rising sovereign yields lift the cost of borrowing for firms, reducing private investment (financial crowding out).
- Uncertainty about future taxation (needed to service debt) reduces investment and labour supply incentives (Ricardian equivalence in weak form).
- Debt sustainability concerns can trigger capital flight and currency depreciation (Argentina, Sri Lanka), worsening macroeconomic conditions.
- High debt limits the fiscal space available to respond to future shocks, making downturns deeper and slower to recover from.

Evaluation — high debt does not necessarily harm growth

- Japan's 252% debt ratio has not caused high yields, high inflation or a debt crisis — context matters (domestic ownership, currency denomination, central bank bond purchases).
- If debt finances productive public investment (infrastructure, R&D, education), it can raise LRAS and therefore long-run growth.
- Keynesian view: deficit spending in a recession can prevent output from falling and preserve the productive capacity of the economy.
- Low real interest rates mean the cost of servicing debt is low — Japan's debt service is manageable despite the high ratio.
- The composition of debt matters: long-dated, domestically-held, domestic-currency debt (Japan, UK) is much less risky than short-term foreign-currency debt (Argentina).
- Advanced economies with reserve currencies (USD, GBP, EUR, JPY) have greater tolerance for high debt.

Possible judgement

- The relationship between debt and growth depends critically on what the debt financed and who holds it.
- There is no magic threshold — the debt-to-GDP ratio alone does not determine harm.
- Debt is less harmful when the economy is below potential (multipliers are high) and when interest rates are low.
- Debt becomes more harmful when debt-service costs rise sharply, when yields rise, or when the debt is in foreign currency.
- The institutional and political context — central bank independence, reserve-currency status, creditor composition — matters more than the debt ratio.

- Long-run growth depends on productivity-enhancing public investment — deficits financing current consumption are more likely to harm long-run growth than those financing capital investment.

Section C — Question 4*

Evaluate whether free trade benefits all countries that participate in it. [25]

Level descriptors as for Q2.*

Indicative content — Question 4*

Knowledge and understanding

- Free trade is international trade without tariffs, quotas or other barriers.
- Comparative advantage: a country gains by specialising in the good for which its opportunity cost is lowest, then trading.
- Trade creation vs trade diversion; static vs dynamic gains from trade.

Application — context

- The World Trade Organization promotes free trade among its 164 member countries.
- Trade liberalisation has been a driver of growth in emerging economies (China, Vietnam, India).
- Some economies remain reliant on primary commodity exports (e.g. Argentina on soybeans, beef, maize).
- Developed and developing countries have rejected trade agreements where benefits are perceived as unequal.

Analysis — free trade benefits all participants

- Comparative advantage: when countries specialise according to opportunity cost, total world output rises and all participants can gain through trade.
- Consumers gain through lower prices and a greater variety of goods, raising real incomes.
- Competition from imports forces domestic firms to be more efficient, raising productivity over time.
- Access to larger markets allows firms to achieve economies of scale, lowering long-run average costs.
- Knowledge and technology transfers through trade accelerate development in lower-income countries.
- Trade enables developing economies with concentrated resource endowments (oil, agricultural commodities) to earn foreign currency.

Evaluation — free trade may not benefit all participants

- Gains from trade are distributed unevenly — some workers, firms and regions lose even as the country as a whole gains (Stolper–Samuelson).
- Infant industry argument: developing countries may need temporary protection before they can compete globally.
- Primary-commodity-dependent economies (e.g. Argentina) face volatile export earnings and the Prebisch–Singer hypothesis of declining terms of trade over time.
- Without safety nets, structural adjustment causes persistent unemployment in declining industries.
- Dumping by foreign producers (e.g. steel below cost) can eliminate domestic production capacity.
- Developing countries may be squeezed into commodity exports and prevented from moving up the value chain.
- Environmental and labour standards may be undercut ("race to the bottom").

Possible judgement

- Free trade can be Pareto-improving in aggregate but rarely benefits every individual, sector or region within each country.
- Whether it benefits all countries depends on the terms of trade, the distribution of gains and complementary policies (education, infrastructure, safety nets).
- Short-run adjustment costs can be large even when long-run gains are positive.
- Developing countries with weak institutions may struggle to capture the benefits of free trade.
- The empirical record is mixed: the East Asian "tigers" benefited from trade, while some African and Latin American economies have not converged.
- A managed, gradual opening with complementary support is often more beneficial than rapid liberalisation.

Section C — Question 5*

Evaluate whether central banks should always prioritise the control of inflation over the pursuit of low unemployment. [25]

Level descriptors as for Q2.*

Indicative content — Question 5*

Knowledge and understanding

- The Phillips curve: a short-run inverse relationship between inflation and unemployment.
- NAIRU (non-accelerating inflation rate of unemployment): the rate of unemployment consistent with stable inflation.
- A dual mandate: a central bank is charged with pursuing both price stability and employment (e.g. Federal Reserve).
- An inflation-only mandate: a central bank's sole objective is inflation (e.g. ECB, Bank of England's primary target).

Application — context

- The Federal Reserve has a dual mandate of price stability and maximum employment.
- The European Central Bank has a primary mandate of price stability (target 2%).
- The Bank of England's primary target is 2% CPI inflation; its secondary objective is to support the government's economic policy including employment.
- Post-pandemic inflation rose above 10% in the UK and eurozone; unemployment has remained relatively low.

Analysis — central banks should always prioritise inflation

- In the long run there is no Phillips curve trade-off — monetary policy can only affect nominal variables (inflation), not real ones (output, unemployment).
- High and volatile inflation erodes real wages, reduces investment through uncertainty and damages the financial system.
- Credibility requires a clear, narrow inflation objective — mandate creep undermines the central bank's anchoring role.
- Unanchored inflation expectations make future policy more costly: the sacrifice ratio rises.
- Hyperinflation episodes (Weimar Germany, Zimbabwe, Argentina) show the catastrophic consequences of not prioritising inflation.
- Low inflation supports long-run growth by reducing menu costs, shoe-leather costs and inflationary tax effects.

Evaluation — central banks should not always prioritise inflation

- When inflation is caused by supply shocks (oil, war, pandemic), tightening policy damages employment without addressing the root cause.
- The welfare cost of high unemployment is severe — job losses cause scarring, skills atrophy and long-term social damage.
- Hysteresis effects mean temporary unemployment can become permanent, reducing LRAS if policy tightens too aggressively.
- A dual-mandate Fed has arguably delivered better average macroeconomic outcomes than a single-mandate ECB (lower average unemployment with comparable inflation).
- In a deep recession, deflation can be as dangerous as inflation — focusing exclusively on an inflation target misses the risk.
- The Phillips curve, while flattened, still implies a short-run trade-off policymakers can exploit during transitions.

Possible judgement

- Whether inflation should always take priority depends on the source of inflation (demand-pull vs cost-push), the level of unemployment and the degree of hysteresis.

- In normal conditions, inflation-targeting delivers good outcomes; in abnormal conditions (pandemic, war, supply shock) a more flexible approach may be warranted.
- A flexible inflation-targeting approach with a medium-term horizon can accommodate both objectives without sacrificing credibility.
- Dual-mandate central banks have flexibility but require exceptionally credible commitment to low inflation to avoid being accused of political bias.
- Ultimately, persistent high inflation is unsustainable and self-defeating — some priority for inflation is non-negotiable, but absolute priority is rarely optimal.
- The central bank's institutional context (independence, mandate, tools) matters as much as its priorities.

ASSESSMENT OBJECTIVES GRID

Question	AO1	AO2	AO3	AO4	Total	Quant.
1(a)		2			2	
1(b)		1(1)			1	(1)
1(c)	2	1(1)			3	(1)
1(d)	2(2)	2(2)			4	(4)
1(e)*	1	1	3	3	8	
1(f)*	1	1	5(2)	5	12	(2)
2*/3*	6(2)	6(2)	6(2)	7(2)	25	(8)
4*/5*	6	6	6	7	25	
TOTAL	**18**	**20**	**20**	**22**	**80**	**(16)**

END OF MARK SCHEME